AGIBOT A3

Platform Time Synchronization Improvement Plan

Offline-compatible chrony discipline for the robot; strict qualification only when coordinating with a separate motion-capture system.

Executive decision	Replace the stock **ntp2rtc.py** system-clock path with continuous chrony discipline, while preserving A3 startup without Internet and preserving the current HDU-to-PHC, inter-board PTP, sensor, and application chain.

Key benefits and the risk of doing nothing

Improvement	Benefit for A3 and ping-pong integration	Risk in the stock path
Running-clock discipline	Chrony retains fractional NTP measurements, filters network delay, and learns oscillator frequency.	ntp2rtc.py updates the RTC but does not continuously discipline the running system clock.
Offline compatibility	A3 applications and internal board/sensor synchronization start even when NTP is unavailable.	No regression is accepted: standalone robot operation must not depend on venue Internet.
Controlled rate	No continuous makestep and maxslewrate 100 ppm reduce disturbance propagated through phc2sys/PTP.	Fast corrections can distort the time rate seen by boards, sensors, recorders, and controllers.
External mocap boundary	A strict preflight is invoked only before the mocap-coordinated task; Motive never controls A3 clocks.	Unqualified clock correlation can bias measurement age and ball-state timing while ROS topics still look healthy.

Protected behavior	Normal A3 startup does **not** contain an NTP precondition. The installed bootstrap service is disabled by default. The production launcher reports UTC quality but always continues to the preserved **ptp4l -i eth_hdu -2 -E** and **phc2sys -s CLOCK_REALTIME -c eth_hdu** path.

Operating rule	Standalone A3: continue when Internet/NTP is absent and mark external UTC unqualified. External mocap: withhold only the mocap-coordinated task until strict UTC and task-latency checks pass. Internal board/sensor health remains a robot requirement in both profiles.

1. Scope, Goals, and Invariants

The design improves the A3 system clock without making Motive or Internet reachability part of the robot's functional dependency chain.

Goals

- Continuously discipline HDU CLOCK_REALTIME with chrony whenever approved NTP sources are reachable.
- Preserve current A3 application startup, inter-board time distribution, and synchronized sensor behavior when NTP is unreachable.
- Keep Motive/NatNet data-only: no path from mocap into chrony, RTC, PHC, ptp4l, or clock_settime.
- Expose a strict, explicit check for the external-mocap task without placing it in agibot_pm startup.

Two operating profiles

Profile	NTP condition	A3 applications	External mocap task
A - Standalone / offline	Optional; may be absent	Start and remain available	Not qualified
B - External mocap	Required to meet approved UTC gates	Already running normally	Start only after separate preflight and latency checks

Non-negotiable invariants

ID	Requirement
I-1	Chrony is the only continuous owner that disciplines HDU CLOCK_REALTIME.
I-2	Normal A3 application startup has no Wants, Requires, or ExecStartPre dependency on NTP qualification.
I-3	No continuous makestep; steps are manual, supervised, and performed with affected applications stopped.
I-4	Preserve CLOCK_REALTIME -> eth_hdu PHC -> internal boards and the active ptp4l E2E mode.
I-5	Continuous correction is capped at maxslewrate 100 ppm.
I-6	An NTP outage marks UTC unqualified but cannot stop standalone A3 operation.
I-7	External mocap qualification cannot alter A3 time ownership or the internal time domain.

Non-goal	This change does not redesign A3's working internal PTP or per-sensor synchronization. Any such redesign is a separate compatibility project.

2. Verified Baseline and Change Boundary

The baseline was inspected read-only. Fleet installation still re-verifies each robot because package state and file ownership can vary by image.

Observed production path

- agibot_pm launches /opt/agibot/entry/system/timesync.sh; that script is the active production owner path.
- The stock script disables other time services, invokes ntp2rtc.py, and launches ptp4l plus phc2sys.

```
ptp4l -i eth_hdu -2 -E -m
phc2sys -s CLOCK_REALTIME -c eth_hdu -w -O 0 -S 10 -m
```

- agibot_timesync.service exists but was disabled/inactive. It must remain disabled to avoid a duplicate owner path.

Why ntp2rtc.py is insufficient for external clock correlation

Behavior	Consequence
Writes RTC periodically	Does not continuously discipline the running system clock used by ROS and applications.
Discards fractional timestamp data	Introduces quantization that is material when estimating measurement age.
No network-delay estimator	Cannot distinguish server offset from asymmetric or variable network delay.
No oscillator-frequency model	Clock drift can accumulate between RTC updates and across long sessions.

Protected vendor behavior

Object	Preserved contract
ptp4l	Interface eth_hdu, layer-2 transport, E2E delay mechanism, current command arguments.
phc2sys	CLOCK_REALTIME remains source and eth_hdu PHC remains sink.
Readiness	LOWER_UP plus ethtool hardware PHC/TX/RX/raw-clock capability checks.
Metadata	Installer records and reapplies the numeric owner, group, and mode of the vendor launcher.

Fleet rule	Detect chrony, ethtool, and linuxptp package presence. Install only genuinely missing dependencies; do not infer package state from one audited A3.

3. Required Architecture and Clock Ownership

Permitted control direction	Approved regional NTP -> chrony -> HDU CLOCK_REALTIME -> vendor phc2sys -> eth_hdu PHC -> vendor ptp4l -> A3 boards/MDU -> existing sensor synchronization and timestamp conversion

Separate observation path	Motive Windows -> NatNet -> ROS 2 adapter -> observations. There is no path from this chain into chrony, CLOCK_REALTIME, RTC, PHC, ptp4l, or sensor clocks.

Layer contracts

Layer	Owner	Contract
UTC reference	Approved NTP profile	Exactly one regional policy; no DHCP-injected sources.
HDU system clock	chrony.service	Slew-only at runtime; no sync prerequisite for A3 apps.
HDU PHC	Vendor phc2sys	CLOCK_REALTIME source; eth_hdu sink; arguments preserved.
Boards	Vendor ptp4l/clients	Existing E2E mode, roles, domain, and lock preserved.
Sensors	Existing firmware/drivers	Existing timestamp point and conversion preserved.
Applications	A3 / ROS 2	Consume documented clock domains; do not own platform time.

Independent health conclusions

Conclusion	What it proves	What it does not prove
A3 internally synchronized	PHC/PTP/boards/sensors meet relative-time contracts.	Absolute UTC agreement with a remote host.
A3 UTC qualified	HDU meets approved external-time source and quality gates.	Internal board/sensor coherence or Wi-Fi latency.
Mocap task qualified	UTC, timestamp semantics, age, jitter, and task gates pass together.	Permission for mocap to discipline A3 clocks.

Normal startup sequence

```
chrony starts if enabled (it does not wait for synchronization)
agibot_pm starts after chrony service ordering only
timesync.sh --runtime-status reports qualified/unqualified and always returns success
ptp4l and phc2sys start with preserved arguments
A3 applications continue whether or not NTP is reachable
```

4. Continuous Chrony Configuration

The continuous profile disciplines the system clock gently and remains usable when all NTP sources are absent.

```
sourcedir /etc/chrony/sources.d
keyfile /etc/chrony/chrony.keys
driftfile /var/lib/chrony/chrony.drift
ntsdumpdir /var/lib/chrony
logdir /var/log/chrony
rtcsync

minsources 2
maxupdateskew 100.0
maxslewrate 100
corrtimeratio 3
logchange 0.1
log tracking statistics

# Absent: makestep, allow, local, sourcedir /run/chrony-dhcp
```

Control rationale

Directive	A3 rationale
minsources 2	One reachable endpoint cannot by itself select the system time.
maxslewrate 100	Bounds the additional correction rate distributed downstream. This is distinct from maxupdateskew.
corrtimeratio 3	Makes correction timing explicit and auditable.
rtcsync	Lets the kernel periodically copy synchronized system time to the RTC when appropriate.
no makestep	Prevents a continuous-service clock step during robot operation.
no /run/chrony-dhcp	Venue DHCP cannot silently bypass approved source governance.
no local	An unsynchronized A3 does not claim to be a valid NTP reference.

Service behavior

- chrony.service has Restart=on-failure, RestartSec=5, and bounded start-rate settings.
- The bootstrap unit is installed for supervised recovery but disabled in normal operation.
- After=agibot-clock-bootstrap.service does not pull in the disabled bootstrap unit.

Rate versus convergence	At 100 ppm, large offsets recover slowly. This is intentional during live operation; use the supervised re-step procedure only when external coordination cannot wait.

5. Package Integration and Startup Semantics

The executable source of truth is the reviewed agibot/ntp_sync package. This PDF deliberately does not duplicate complete scripts.

Canonical engineering files

Package path	Purpose
README.md	Self-contained quick installation for normal offline-capable A3 operation.
opt/agibot/entry/system/timesync.sh	Reviewed vendor-launcher replacement; non-blocking runtime status plus explicit strict preflight.
etc/chrony/chrony.conf	Continuous, slew-only system-clock discipline.
etc/systemd/system/agibot_pm.service.d/20-clock-ordering.conf	After=chrony.service only; no startup preflight and no dependency pull-in.
etc/systemd/system/chrony.service.d/agibot-controls.conf	Failure restart and ordering controls.
usr/local/sbin/agibot-time-region	Atomic selection of one approved regional profile while applications are stopped.
usr/local/sbin/agibot-clock-bootstrap	Manual supervised one-shot correction; disabled by default.
MANIFEST.sha256	Integrity check for all package artifacts.

timesync.sh interfaces

Invocation	Return policy	Use
normal (no option)	Does not fail because chrony/NTP is unavailable	Production A3 internal time distribution.
--runtime-status	Always returns success after reporting clock state	Normal operations and quick-guide verification.
--preflight	Fails closed unless strict UTC checks pass or an approved override is active	External-mocap commissioning only.

agibot_pm drop-in

```
[Unit]
After=chrony.service
```

- No Wants= or Requires=: chrony is not pulled in as a functional dependency by the robot application.
- No ExecStartPre=: NTP reachability and UTC qualification cannot block A3 startup.
- Ordering avoids a harmless early warning race when chrony is enabled.

Document control	Review and deploy complete files from agibot/ntp_sync after verifying MANIFEST.sha256. Do not reconstruct executable artifacts from PDF excerpts or historical package copies.

6. Installation Contract and Normal Verification

The quick README is the operational entry point. This section defines what its successful completion must mean.

Maintenance constraints

- Robot supported; motion disabled; external commands and recording stopped.
- Record the currently active A3 services and restore only that same set.
- Stop agibot_pm and existing time processes before changing the clock owner.
- Preserve vendor-launcher numeric ownership, group, and mode; verify shell syntax and chrony parsing before restart.
- Reject any extra .sources file; select exactly one regional profile.

Normal activation

```
systemctl disable --now agibot-clock-bootstrap.service
systemctl unmask chrony.service
systemctl enable --now chrony.service
# waitsync is informative and non-blocking in the README
/opt/agibot/entry/system/timesync.sh --runtime-status
# restore only A3 services that were active before maintenance
```

Acceptance at the end of the README

Check	Required result
Clock owner	Exactly one chronyd; no ntp2rtc.py and no second NTP daemon.
Internal PTP	ptp4l still has -i eth_hdu -2 -E.
PHC direction	phc2sys still has -s CLOCK_REALTIME -c eth_hdu.
Online	chronyc shows Leap status Normal and a selected ^* source after convergence.
Offline	A3 applications remain active; runtime status reports UTC unqualified; strict mocap use is withheld.
Bootstrap	agibot-clock-bootstrap.service remains disabled.

Single retry is not a fault	On an online boot, agibot_pm may start before chrony has selected a source. The runtime check may initially report UTC unqualified, but it does not block or restart the robot chain.

7. Offline Operation and Fault Isolation

Internet loss is a clock-quality event, not a loss of standalone robot function.

Expected cold boot with no Internet

Component	Expected state
chrony.service	Active but unsynchronized; may retain learned frequency correction from its drift file.
CLOCK_REALTIME	Starts from the platform's normal RTC/boot path; not stepped by the disabled bootstrap service.
agibot_pm and applications	Start according to their existing enablement and dependencies.
ptp4l / phc2sys	Start from the vendor launcher and maintain the existing internal time domain.
ROS 2	Continues using A3 time; absolute correlation to remote hosts is unqualified.
External mocap	Do not start coordinated operation until qualification is restored.

Fault matrix

Scenario	Required behavior	Scope of inhibition
DNS failure / UDP 123 blocked	chrony remains active; A3 continues	External mocap task only
All regional sources unavailable	UTC unqualified warning; no clock step	External mocap task only
chronyd crash	systemd restarts it; launcher continues internal sync	Alarm; do not stop standalone A3 solely for UTC
NTP returns	chrony slews at <=100 ppm; no runtime step	Mocap remains unqualified until strict checks pass
Motive disconnect/restart	No A3 clock-owner or PTP change	Mocap task
Board/sensor sync failure	Existing platform fault response	Robot motion according to vendor safety policy

Offline acceptance exercise

- On a supported robot, cold boot with the external Internet unavailable but the internal A3 network intact.
- Confirm enabled A3 services start; verify exact ptp4l/phc2sys arguments and absence of ntp2rtc.py.
- Confirm chronyc reports unsynchronized without causing an agibot_pm start failure.
- Restore Internet and confirm chrony converges by slew; no application restart is required.

Do not disable eth_hdu	The offline test removes only external NTP reachability. It must not disrupt the wired internal board/sensor network used by ptp4l and phc2sys.

8. External Mocap Qualification

This gate is applied to the mocap-coordinated ping-pong workflow, not to A3's base application service.

Entry conditions

- Normal A3 operation and internal board/sensor synchronization are healthy.
- The approved regional profile is active; no injected chrony sources exist.
- chrony has converged and the strict external-time preflight succeeds.

```
chronyc waitsync 60 0.010 5 2
/opt/agibot/entry/system/timesync.sh --preflight
chronyc tracking
chronyc sources -v
chronyc sourcestats -v
```

- Replace the illustrative 10 ms / 5 ppm limits with owner-approved values after venue measurement.

Task-level gates

Gate	Evidence	Failure action
A3 UTC	Selected source, Leap Normal, correction/skew/dispersion within approved limits	Do not start or pause external mocap coordination
Internal A3	PHC/board/sensor offset, lock, role, monotonicity within owner limits	Follow A3 safety policy; mocap qualification cannot waive this
NatNet semantics	Native frame/timestamps retained; mapped time domain documented	Reject or fall back from time-sensitive mocap use
Network age/jitter	Venue distribution of capture-to-consume age, p95/p99/max, loss and reordering	Tune or inhibit the mocap-coordinated task
Controller timing	No deadline bursts, stale-state use, or backward timestamps	Stop the task and retain evidence

Approved holdover override

- AGIBOT_ALLOW_UNQUALIFIED_TIME=1 affects only the explicit --preflight command. Normal A3 operation does not need this override.
- Use only with two-person sign-off, a measured holdover envelope, and visible UTC-unqualified status.
- Restore the value to 0 after the approved window. It must never become a standing way to bypass external-mocap qualification.

Important	Common NTP sources for Windows and A3 can aid diagnosis, but they do not by themselves prove Motive capture-time alignment or remove Wi-Fi delay uncertainty.

8.1 Holdover Activation Command Runbook

Holdover is an explicitly approved, UTC-unqualified mode for the external-mocap task. Standalone A3 operation never needs this override.

Before activation	Obtain two-person approval; record the maximum duration and drift budget; stop the external mocap task; confirm A3 internal board/sensor synchronization is healthy. Holdover does not claim UTC accuracy.

Activate the override and visible operator marker

```
sudo systemctl is-active chrony.service
sudo /usr/local/sbin/agibot-time-region --show

sudo sed -i -E \
's/^AGIBOT_ALLOW_UNQUALIFIED_TIME=.*/AGIBOT_ALLOW_UNQUALIFIED_TIME=1/' \
/etc/default/agibot-timesync
sudo chown root:root /etc/default/agibot-timesync
sudo chmod 0644 /etc/default/agibot-timesync
grep '^AGIBOT_ALLOW_UNQUALIFIED_TIME=1$' /etc/default/agibot-timesync

sudo install -d -o root -g root -m 0755 /var/lib/agibot-time
printf 'UTC_UNQUALIFIED holdover enabled %s\n' "$(date -u +%FT%TZ)" | \
sudo tee /var/lib/agibot-time/utc-unqualified-holdover >/dev/null
sudo chmod 0644 /var/lib/agibot-time/utc-unqualified-holdover

if ! sudo /opt/agibot/entry/system/timesync.sh --preflight; then
echo 'Holdover activation failed; restoring fail-closed policy.'
sudo sed -i -E \
's/^AGIBOT_ALLOW_UNQUALIFIED_TIME=.*/AGIBOT_ALLOW_UNQUALIFIED_TIME=0/' \
/etc/default/agibot-timesync
sudo rm -f /var/lib/agibot-time/utc-unqualified-holdover
exit 1
fi
chronyc tracking
```

Required result

- The explicit preflight succeeds only because the signed override is active; logs and operator UI must continue to say UTC_UNQUALIFIED.
- Keep monitoring elapsed holdover time, chrony state, PHC/PTP offsets, boards, sensors, and application timestamp behavior.
- Do not use this override to hide failed source governance, an inactive chrony daemon, or failed internal synchronization.

8.2 Holdover Exit and Requalification Commands

Close holdover as soon as approved NTP returns. The policy is restored to fail closed before attempting external-time requalification.

Before exit	Stop the external mocap task. Standalone A3 applications may continue because this procedure changes only the explicit external-time preflight policy.

```
sudo sed -i -E \
's/^AGIBOT_ALLOW_UNQUALIFIED_TIME=.*/AGIBOT_ALLOW_UNQUALIFIED_TIME=0/' \
/etc/default/agibot-timesync
sudo chown root:root /etc/default/agibot-timesync
sudo chmod 0644 /etc/default/agibot-timesync
grep '^AGIBOT_ALLOW_UNQUALIFIED_TIME=0$' /etc/default/agibot-timesync

chronyc online
chronyc refresh
if chronyc waitsync 60 0.010 5 2 && \
sudo /opt/agibot/entry/system/timesync.sh --preflight; then
sudo rm -f /var/lib/agibot-time/utc-unqualified-holdover
echo 'UTC requalified; proceed to board/sensor and mocap task gates.'
else
printf 'UTC_UNQUALIFIED holdover closed; requalification failed %s\n' \
"$(date -u +%FT%TZ)" | \
sudo tee /var/lib/agibot-time/utc-unqualified-holdover >/dev/null
echo 'Do not restart external mocap coordination.'
exit 1
fi
chronyc tracking
chronyc sources -v
chronyc sourcestats -v
```

If correction is still large

- The waitsync command requires less than 10 ms remaining correction and less than 5 ppm skew; replace these examples with owner-approved limits.
- At the 100 ppm runtime cap, a large remaining correction may take too long. Use the supervised re-step command runbook instead of raising the live slew rate.

Exit state	The override remains 0 even when requalification fails. Standalone A3 remains available, but external mocap stays prohibited and the marker records the unresolved UTC-unqualified state.

9. Supervised Re-step and Recovery Timing

Continuous slew protects a live robot but may be too slow after a badly wrong RTC. Recovery is an explicit maintenance action, never an automatic boot dependency.

100 ppm recovery bound

Initial offset	Best-case correction time	External mocap disposition
10 ms	100 seconds	Wait and requalify
0.3 s	50 minutes	Operator decides whether to wait or re-step
2 s	5 hours 33 minutes	Normally use supervised re-step
30 s	83 hours 20 minutes	Supervised re-step required for timely recovery

Supervised procedure

- Support the robot, disable motion, stop recording/external commands, and obtain two-person approval.
- Stop agibot_pm and the active A3 application services; stop ptp4l, phc2sys, and chrony.
- Select and validate the approved regional profile. Confirm NTP reachability and expected offset before changing the clock.
- Run the one-shot agibot-clock-bootstrap unit manually. Its script creates /run markers before any optional RTC write.
- Start continuous chrony, wait for strict qualification, then restart only services recorded as active.
- Re-run internal board/sensor and external mocap checks before enabling the mocap-coordinated task.

Bootstrap controls

Control	Requirement
Default state	Disabled; never part of ordinary offline boot.
Timeout	BOOTSTRAP_TIMEOUT_S in 5..150 seconds; systemd outer ceiling 180 seconds.
Step window	Only after affected application and distribution processes are stopped.
Markers	bootstrap-qualified records successful one-shot correction; rtc-updated separately records RTC write.
Failure	Keep A3 in the maintenance state for external coordination; normal standalone profile can be restored without enabling bootstrap.

No automatic runtime re-step	Do not add makestep to continuous chrony or leave the bootstrap unit enabled for ordinary unattended boots. A controlled online boot or supervised manual bootstrap is allowed only while protected services are ordered or stopped, followed by immediate bootstrap disablement.

9.1 Complete Supervised Re-step Command Runbook

Use this when a large startup error cannot converge by 100 ppm slew before external mocap is needed. The commands restore standalone A3 even when external-time qualification fails.

Physical state	Support the robot; disable motion, recording, and external commands; confirm the approved regional NTP profile and network are available. Obtain two-person approval before the clock step.

```
RESTEP_STATE=/tmp/agibot-restep-active-services
: > "$RESTEP_STATE"
for SERVICE in agibot_roudi agibot_top agibot_ui agibot_pm; do
systemctl is-active --quiet "$SERVICE" && \
printf '%s\n' "$SERVICE" >> "$RESTEP_STATE"
done

sudo systemctl stop agibot_pm || true
sudo systemctl stop agibot_ui agibot_top agibot_roudi || true
sudo pkill -TERM -x phc2sys || true
sudo pkill -TERM -x ptp4l || true
sudo systemctl disable agibot-clock-bootstrap.service
sudo systemctl stop chrony.service
sleep 2

if pgrep -a -f '[c]hronyd|[p]tp4l|[p]hc2sys'; then
echo 'STOP: a protected time process is still active'
exit 1
fi

MOCAP_TIME_READY=0
if sudo systemctl restart agibot-clock-bootstrap.service && \
sudo test -e /run/agibot-time/bootstrap-qualified; then
sudo systemctl start chrony.service
if chronyc waitsync 60 0.010 5 2 && \
sudo /opt/agibot/entry/system/timesync.sh --preflight; then
MOCAP_TIME_READY=1
fi
else
echo 'Re-step failed; restoring standalone A3 with UTC unqualified.'
sudo systemctl start chrony.service
fi

while IFS= read -r SERVICE; do
sudo systemctl start "$SERVICE"
done < "$RESTEP_STATE"
printf 'MOCAP_TIME_READY=%s\n' "$MOCAP_TIME_READY"
```

Interpretation

- MOCAP_TIME_READY=1 qualifies only the A3 external-time layer. Verify ptp4l/phc2sys arguments, internal boards/sensors, NatNet semantics, and task latency before play.
- MOCAP_TIME_READY=0 prohibits external mocap coordination but does not prohibit normal standalone A3 operation.
- The bootstrap service remains disabled after this manual invocation; continuous chrony returns to the 100 ppm runtime cap.

10. Board and Sensor Preservation Tests

The statement that A3 boards and sensors are synchronized is a protected platform contract. The chrony change must demonstrate non-regression, not replace that mechanism.

Inventory template

Object	Timestamp point and path	Owner limits / failure action
HDU / MDU / boards	PHC, system clock or counter; PTP role/domain/offset	Offset, jitter, lock, role change
Cameras / vision	Exposure/frame/driver timestamp; trigger/PTP/mapping	Age, drops, reset, alignment
IMU	Sample counter and driver stamp	Monotonicity, drift, jitter
Force / contact	Sample or bus cycle time	Cycle alignment, timeout
Joint / motor feedback	Board timestamp and bus cycle	Period, offset, duplicate/backward samples

Comparison protocol

- Collect at least 30 minutes before and after under equivalent CPU, network, sensor, and recording load.
- Report median, p95, p99, maximum offset and jitter, dropouts, role changes, resets, and discontinuities.
- Exercise online boot, offline boot, NTP loss/return, chronyd crash, PTP client disconnect, sensor reset, and supervised re-step.
- Confirm no change in ptp4l delay mechanism, domain, transport, grandmaster role, or phc2sys direction.

Decision table

A3 internal sync	External UTC	Allowed operation
Pass	Pass	Standalone A3 and qualified external-mocap task
Pass	Fail / unavailable	Standalone A3; external mocap task withheld
Fail	Pass	Follow A3 platform safety policy; UTC does not compensate
Fail	Fail	Follow A3 platform safety policy; investigate both domains

Causality	Clock-domain error is a credible pathway to stale or misordered observations, but it is not proof of a specific balance loss. Correlate clock traces with controller deadlines, commanded targets, contact events, and fall logs.

11. Motive, NatNet, and ROS 2 Time Boundary

Motive is an observation producer. External time mapping belongs in the integration layer and does not discipline the A3 platform.

Timestamp contract

Field	Meaning
motive_frame_number	Native frame identity for loss, duplicate, and reordering detection.
TransmitTimestamp	Motive/NatNet server transmit point in the server high-resolution counter domain.
CameraMidExposureTimestamp	Preferred capture point when supplied by the camera system.
receive_timestamp	Relay/adapter receive time for transport age and jitter measurement.
header.stamp	Documented consumer-domain timestamp; never silently relabel a Motive counter as A3 wall time.
capture_time_estimate	Optional affine mapping with reported uncertainty and freshness.

Estimator placement

- Estimate t_A3 = a * t_Motive + b from native performance-counter timestamps and paired receive observations.
- Place the estimator on the wired relay host nearest Motive when possible; Wi-Fi delay and asymmetry should remain downstream measurements, not estimator inputs that are mistaken for clock offset.
- Reject stale estimates after Motive restart, counter discontinuity, network outage, or uncertainty growth.

Venue latency test

Measure	Report
Capture-to-transmit	Native Motive metadata when available; frame identity and counter continuity.
Transmit-to-relay receive	Median, p95, p99, max; packet loss and reordering.
Relay-to-A3 consume	Wi-Fi/ROS path age and jitter under representative traffic.
End-to-end observation age	Distribution at controller consumption, not only average network ping.
Clock-map uncertainty	Estimator residual, age, drift, reset detection, and confidence bound.

Isolation test	Disconnect and restart Motive/NatNet. Prove that chronyd, CLOCK_REALTIME rate, phc2sys, ptp4l, board lock, and sensor synchronization do not change.

12. Monitoring, Rollout, and Rollback

Monitoring by domain

Domain	Monitor
NTP / HDU	tracking, sources, sourcestats, selected source, correction, skew, dispersion, restart count
PHC	phc2sys arguments, offset/frequency/delay distribution
PTP boards	roles, role changes, delay mechanism, domain, offset, reachability, duplicate owners
Sensors	native state, counters, monotonicity, offset/jitter/age, drops and resets
Applications	backward/duplicate stamps, timeout bursts, stale-state use, control anomalies
Mocap boundary	frame loss/reordering, native counter resets, mapping residual, observation age

Rollout

Phase	Change	Exit evidence
0	Inventory and baseline	Package state, active services, owners, board/sensor mechanisms and limits recorded
1A	Install in supported maintenance state	Config parses; manifest passes; bootstrap disabled; no duplicate owners
1B	Bench online and offline	A3 starts in both; exact internal time commands preserved; no regression
1C	External mocap rehearsal	Strict UTC, NatNet semantics, latency/age, and controller gates pass
1D	Production	Cold boots, soak, rollback drill, owner sign-off complete

Rollback triggers

- Any board/sensor regression, clock step, rate-cap violation, duplicate owner, role/direction change, application startup regression, or source-governance failure.

Rollback principles

- Use the recorded backup and service-enable state. Stop applications and time processes before restoration.
- Restore vendor timesync.sh with captured metadata and restore the prior chrony.conf if one existed.
- Remove only files introduced by this package, reload systemd, and restart only services that were previously active/enabled.
- Verify stock ptp4l/phc2sys behavior. Mark external UTC correlation unqualified after rollback.

Operational meaning	Rollback restores prior A3 behavior, including offline operation, but it also restores the stock system-clock limitations. External mocap must remain unqualified until another approved clock solution is commissioned.

13. Regional NTP Governance and Leap Policy

Region	Configured public endpoints	Authority
China (default)	ntp1.aliyun.com; ntp2.aliyun.com; ntp3.aliyun.com	Alibaba Cloud public NTP, IPv4
US	time-a-g.nist.gov; time-b-g.nist.gov; time-c-g.nist.gov	NIST Internet Time Service
Europe	ptbtime1.ptb.de; ptbtime2.ptb.de; ptbtime3.ptb.de	PTB public time service

Governance

- China is the default. Switch regions only while agibot_pm is stopped; exactly one hope-approved.sources link is allowed.
- Each current profile uses one provider. A provider or routing outage leaves A3 operational but UTC unqualified for external coordination.
- Public NTP is unauthenticated in these profiles. Approved endpoints do not eliminate the network trust boundary.
- Do not mix known leap-smearing and standard-UTC providers in one active profile.
- leapsectz is omitted because regional source behavior differs and right/UTC may not be installed. Revalidate provider leap policy before a scheduled leap event.

Official references

- Alibaba Cloud public NTP server list
- NIST Internet Time Service server list
- PTB NTP and NTS service
- chrony 4.3 configuration reference
- chronyc 4.3 command reference
- LinuxPTP phc2sys documentation and ptp4l documentation

Source review date	Endpoint lists and documentation were rechecked on 2026-08-04. Reconfirm them during venue commissioning because public-service policies can change.

14. Authorization and Review Record

v1.7 disposition

Issue	Disposition
Offline boot regression	Resolved: no NTP preflight in agibot_pm; normal runtime status always returns success.
Bootstrap at every boot	Resolved: installed for supervised recovery but disabled by default.
Mocap dependence	Resolved: strict UTC qualification applies only to the external-mocap task.
Internal board/sensor chain	Preserved: existing ptp4l/phc2sys direction and readiness checks remain.
Executable drift in PDF	Resolved: agibot/ntp_sync is the sole canonical code package; PDF documents contracts and procedures.
Unapproved source injection	Controlled: /run/chrony-dhcp omitted; exactly one approved .sources file required.
Runtime correction rate	Controlled: maxslewrate 100, corrtimeratio 3, no continuous makestep.
Incident recovery commands	Resolved: complete holdover activation/exit and supervised re-step command blocks are included in Sections 8.1, 8.2, and 9.1.

Required signatures before production external-mocap use

Role	Name / signature	Date	Decision
Robot controls			GO / NO-GO
Board / firmware			GO / NO-GO
Sensor integration			GO / NO-GO
Platform			GO / NO-GO
Mocap / ROS 2			GO / NO-GO
Event operations			GO / NO-GO

Final acceptance	Normal A3 operation must pass both online and offline startup tests before deployment. External-mocap authorization additionally requires owner-approved clock, board/sensor, NatNet timestamp, observation-age, and controller-timing limits.

Document control

- Version: Authoritative v1.7
- Canonical executable package: agibot/ntp_sync/
- Operational quick start: agibot/ntp_sync/README.md
- Supersedes all v1.0-v1.6 English copies and any historical generated package locations.